Irina Saranskaia

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• irinasaranskaia.info •

Research interests: speech processing, predictive coding, neuroimaging analysis

Education

Institute for Cognitive Neurosciences, **HSE University**
PhD in Cognitive Sciences
Moscow, Russia, **2025 – present**

Institute for Cognitive Neurosciences, **HSE University**
M.S., Cognitive Sciences and Technologies
Moscow, Russia, **2023 – 2025**
Graduated with excellence

Liberal Arts and Sciences, **St. Petersburg State University**
B.S., Computer Science and Artificial Intelligence
St. Petersburg, Russia, **2019 – 2023**
Graduated with excellence

Publications

Vlasenko, D., **Saranskaia, I.**, & Zakharov, D. (2026). Hypergraphs from multivariate connectivity: caCoh-based EEG/MEG representation. *European Physical Journal Special Topics*. https://doi.org/10.1140/epjs/s11734-026-02558-7

Vlasenko, D., **Saranskaia, I.**, & Zakharov, D. (2025). From pairwise to high-order: Hypergraph methods and multivariate connectivity metrics for EEG/MEG. In *Proceedings of the 2025 7th International Conference Neurotechnologies and Neurointerfaces (CNN)*, 142–145, doi.org/10.1109/CNN67635.2025.11177482

Saranskaia, I., Gutkin, B., & Zakharov, D. (2025) Aim-based choice of strategy for MEG-based brain state classification. *European Physical Journal Special Topics*, 234, 4159–4177. doi.org/10.1140/epjs/s11734-025-01587-y

Research experience

HSE University, Institute for Cognitive Neurosciences
Project "AI-based listening diagnostic systems"
Research assistant, **February 2025 – present**

- Full EEG experiment pipeline execution: experimental design, stimulus set construction, pilot validation, large-scale data acquisition
- Data quality control, trial validation, speech recordings analysis
- EEG preprocessing, time-frequency and spectral analyses, feature analysis
- Scientific writing

Technologies: Python: psycopy, mne, numpy, pandas, scipy, scikit-learn, seaborn and matplotlib, speech/behavioural tooling (JiWER for WER/CER; audio processing with librosa/pyloudnorm/parselmouth/soundfile), Hilbert/PSD/Welch methods; MATLAB: fieldtrip; Praat scripting and .TextGrid annotation

Russian Science Foundation
Grant 24-68-00030 "Next Generation Cognitive Artificial Intelligence"
Research-assistant, **December 2024 – present**

- EEG/MEG data analysis and feature extraction for graph representation for a classification of brain states

Technologies: Python: numpy, scipy, scikit-learn, pandas, matplotlib, mne, mne-connectivity, jpype, infodynamics; MATLAB: fieldtrip.

HSE University
2023 – 2025

- Master's research dedicated to the MEG-based functional connectivity analysis in anxiety patients utilising Time-Reversed Granger causality for functional

connectivity assessment.

Technologies: MEG connectivity analysis, time-reversed Granger causality; python, numpy, pandas, scipy, mne, scikit-learn; MATLAB: eeglab; inverse modelling, LCMV beamforming, statistical analysis, multiple comparison corrections.

St. Petersburg State University
2022 – 2023

- Bachelor's thesis "Reproducibility Study of Different Methods For Brain Networks Comparisons." The thesis served the purpose of formulating recommendations for the use of various correction methods for multiple comparisons.

Technologies: coherency-based EEG analysis; python, numpy, pandas, scipy, multipy; graph-theory, statistical analysis, nonparametric permutation testing, multiple comparison corrections.

Conferences

Paris Île-de-France Neuroscience, Theory, and Systems (PINTS) 2026.
"*Lexical predictability and δ-phase tracking of time-compressed speech*" (poster presentation)
Paris, France. **6 November 2026**

Architectures and Mechanisms for Language Processing (AMLaP) 2026.
"*Lexical Predictability and Speech–Brain Synchrony in Compressed Speech*" (poster presentation)
Saarbrücken, Germany. **2 – 4 September 2026**

XI International Conference on Cognitive Science.
"*Functional brain connectivity and anxiety: MEG during decision-making*"
"*Pupil size and response accuracy in perceptual decision-making in an ambiguous image classification*" (co-authored with N. Istomina)
Nizhny Novgorod, Russia. **24 – 28 August 2026**

NeuroAI Meeting 2025.
"*ML-based approaches for EEG/MEG classification*" (oral presentation).
Moscow, Russia. **24 – 25 November 2025**

VII International Conference "Neurotechnologies and Neurointerfaces" as part of the V International Conference "Volga Neuroscience Meeting 2025".
"*From Pairwise to High-Order: Hypergraph Methods and Multivariate Connectivity Metrics for EEG/MEG*" (poster presentation)
Nizhny Novgorod, Russia. **25 – 29 August 2025**

XXXI Scientific School "Nonlinear Days-2025".
"*Aim-based choice of strategy for MEG-based brain state classification*" (poster presentation).
Saratov, Russia. **26 – 30 May 2025**

XXI Scientific School "Nonlinear Waves-2024".
"*Methods for assessing functional connectivity based on electromagnetic data of brain activity*" (poster presentation).
Nizhny Novgorod, Russia. **5 – 11 November 2024**

Scholarships

Ostrogradski Fellowship**, Ambassade de France en Fédération de Russie**
September – December 2026

Brain Programme Fellowship, **Research Center "IDEAS"**
January 2026 – present

Academic Scholarship**, HSE University**
November 2025 – present

Additional Education

MEG-OPM Toolkit, **Donders Centre for Cognitive Neuroimaging (Donders Institute for Brain, Cognition and Behaviour)**
Nijmegen, the Netherlands. **12–16 October 2026**

Fall School in Computational Neuroscience, **EITN EBRAINS**
Attended with my husband; both independently selected
Pavia, Italy. **21–30 September 2026**

Paris Circuit Dynamics: Neuromodulation of Neuronal Circuits: From Behavior to Computation (Conference & Workshop), **Paris Brain Institute**
Paris, France. **14–17 September 2026**

Professional Development Programme "*Methods of psycho- and neurolinguistics*", **HSE University**
34 academic hours; professional development certificate awarded
St. Petersburg, Russia. **22–24 June 2026**

XIII Summer Neurolinguistics School "*Clinical applications of neurolinguistics in neurosurgery*", **Center for Language and Brain, HSE University**
Moscow, Russia. **1–3 June 2026**

Summer School in NeuroAI, **Neuromatch Academy**
Online. **14 – 25 July 2025**

Summer School in Computational Neuroscience, **Neuromatch Academy**
Online. **8 – 26 July 2024**
Project "Neural correlates of visual processing and action initiation in mouse decision-making"

Languages

Russian (native), English (C1), French (A2)

Skills

Data analysis & machine learning | Analysis of quantitative and categorical data; network analysis; machine (deep) learning; EEG/MEG (ERP/ERF, time-frequency, connectivity analyses) and neural electrode data analysis

Neuroimaging experiment | EEG (BrainVision), EyeTracking (EyeLink)

Speech and audio processing
Analysis and preprocessing of speech data (Praat scripting for segmentation, annotation, and acoustic feature extraction);
Python-based audio libraries for speech and sound analysis: pyloudnorm, librose, parselmouth, signal.

Programming & tools | Python: numpy, pandas, scipy,, scikit-learn, pytorch; nilearn, networkx, mne, infodynamics; Git; Linux/Bash; HTML, CSS, LaTeX; PsychoPy (including custom experiment scripting);
MATLAB: fieldtrip, eeglab.

Mathematics | statistics, probability theory, graph theory, linear algebra, calculus.

References

Dr. Denis Zakharov
Director, Theoretical Neuroscience Group, Centre for Cognition & Decision Making, HSE University, Moscow, Russia
Leading Research Fellow, Centre for Cognition & Decision Making, HSE University, Moscow, Russia
dgzakharov@hse.ru

Dr. Olesia Dogonasheva
Postdoctoral Researcher, Institut de l'Audition (Institut Pasteur / Inserm / CNRS / Université Paris Cité / AP-HP), IHU reConnect, Paris, France
odogonasheva@gmail.com